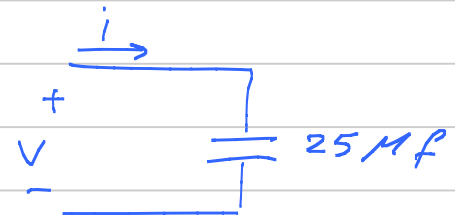
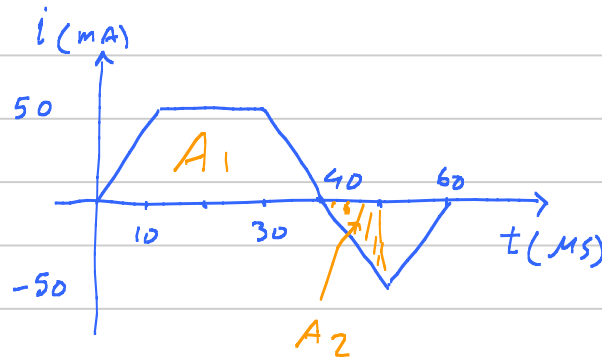




Ex. The current pulse is applied to a 25 μf capacitor with no initial voltage.

- Find the capacitor voltage at $t=50 \mu\text{s}$.
- How much energy is stored in the capacitor by this pulse.



$$\begin{aligned}
 a) \quad v &= \frac{1}{C} \int_0^{50 \text{ ms}} i \, dt + 0 \\
 &= \frac{1}{25 \times 10^{-6}} \left[\underbrace{\frac{1}{2} (20 + 40) \times 10 \times 50 \times 10^{-3}}_{A_1} - \underbrace{\frac{1}{2} (10 \times 10) \times 50 \times 10^{-3}}_{A_2} \right]
 \end{aligned}$$

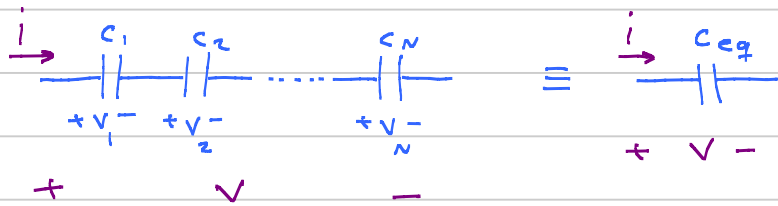
$$V_{50 \text{ ms}} = \underline{50 \text{ mV}}$$

$$b) \quad W_{(60 \text{ ms})} = \frac{1}{2} C V_{(60 \text{ ms})}^2$$

$$V_{(60 \text{ ms})} = \frac{1}{25 \times 10^{-6}} [A_1 - 2A_2] = 40 \text{ mV}$$

$$W = \frac{1}{2} (25 \times 10^{-6}) (40 \times 10^{-3})^2 = \underline{20 \text{ nJ}}$$

series and parallel combination of capacitors

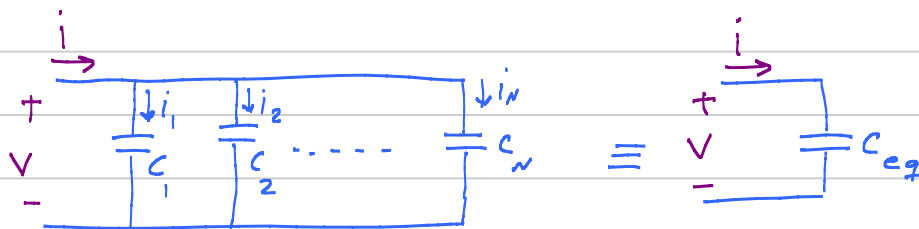


$$V = V_1 + V_2 + \dots + V_N = \frac{1}{C_1} \int i dt + V_{o1} + \frac{1}{C_2} \int i dt + V_{o2} + \dots + \frac{1}{C_N} \int i dt + V_{oN}$$

$$= \left(\frac{1}{C_1} + \frac{1}{C_2} + \dots + \frac{1}{C_N} \right) \int i dt + (V_{o1} + V_{o2} + \dots + V_{oN})$$

$$\Rightarrow \boxed{\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \dots + \frac{1}{C_N}}, \quad V_{o eq} = V_{o1} + V_{o2} + \dots + V_{oN} \quad \text{Series capacitors}$$

For two series capacitors: $C_{eq} = \frac{C_1 C_2}{C_1 + C_2}, \quad V_{o eq} = V_{o1} + V_{o2}$

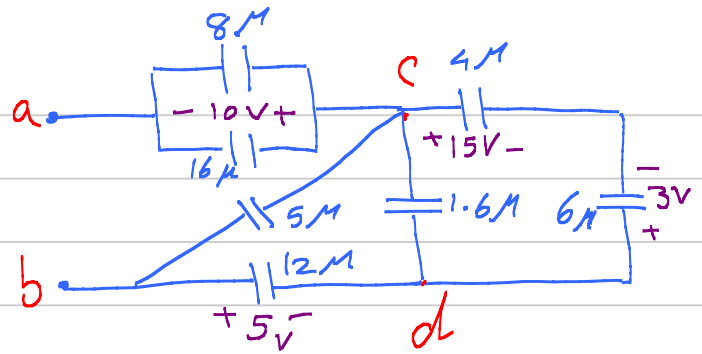


$$i = i_1 + i_2 + \dots + i_N = C_1 \frac{dV}{dt} + C_2 \frac{dV}{dt} + \dots + C_N \frac{dV}{dt}$$

$$= (C_1 + C_2 + \dots + C_N) \frac{dV}{dt}$$

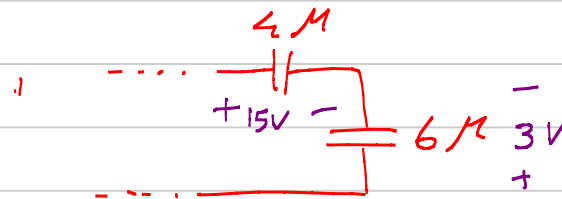
$$\Rightarrow \boxed{C_{eq} = C_1 + C_2 + \dots + C_N} \quad \text{Parallel capacitors}$$

Ex. Find the equivalent capacitance.

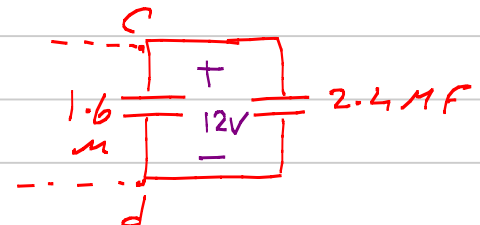


$$\frac{4 \times 6}{4 + 6} = 2.4 \text{ MF}$$

$$V_0 = 15 - 3 = 12 \text{ V}$$

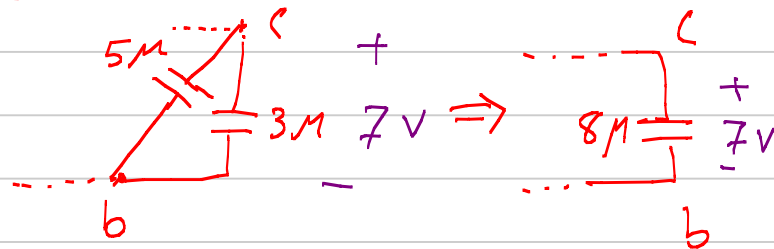


$$1.6 + 2.4 = 4 \text{ MF}, V_0 = 12 \text{ V}$$

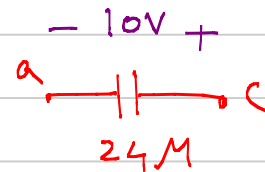


$$\frac{4 \times 12}{4 + 12} = 3 \text{ MF}, V_0 = 12 - 5 = 7 \text{ V}$$

$$5 + 3 = 8 \text{ MF}, V_0 = 7 \text{ V}$$

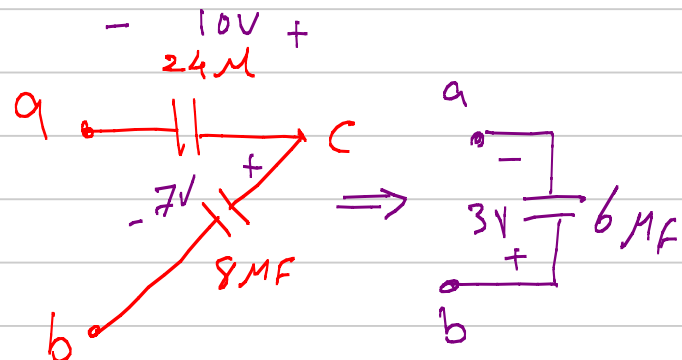


$$8 + 16 = 24 \text{ MF}, V_0 = -10 \text{ V}$$



$$\frac{24 \times 8}{24 + 8} = \boxed{6 \text{ MF}}$$

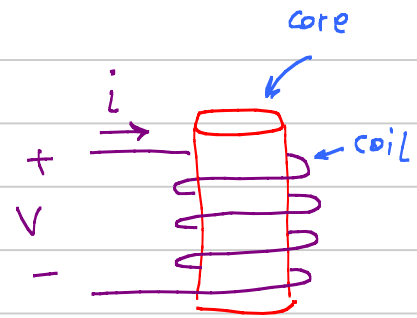
$$V_0 = -10 + 7 = \boxed{-3 \text{ V}}$$



The Inductor :

An inductor is an energy storage element composed of a coil of wire wound around a supporting core. The behavior of inductor is based on magnetic field which is created by charge in motion (current). A time varying current creates a time varying magnetic field which induces voltage in any conductor linked by the magnetic field. The parameter that relates the induced voltage to the current is called inductance, denoted by L and measured in Henrys (H).

$$L = \frac{N^2 \mu A}{l}$$

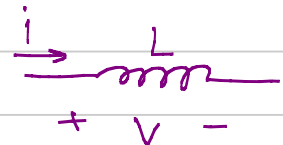


A : Cross sectional area of the core

N : number of turns

l : length of the core

μ : permeability of the core



$$V = L \frac{di}{dt} \quad (\text{if } i \text{ is in the direction of voltage drop})$$

current in terms of voltage

$$i = \frac{1}{L} \int_{t_0}^t v dt + i(t_0) \quad (\text{if } i \text{ is in the direction of voltage drop})$$

Power and Energy

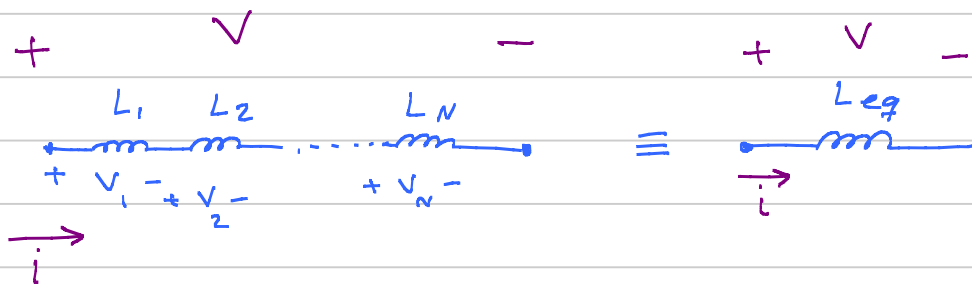
$$P = vi = L i \frac{di}{dt}$$

$$w(t) = \int_{t_0}^t P dt = \int_{t_0}^t L i \frac{di}{dt} dt = \frac{1}{2} L i^2 \Big|_{i(t_0)}^{i(t)} = \frac{1}{2} L i^2(t)$$

Important properties of inductors

1. An inductor behaves as a short circuit when its current is constant.
2. The current in an inductor cannot change instantaneously. (Inductor current is continuous).

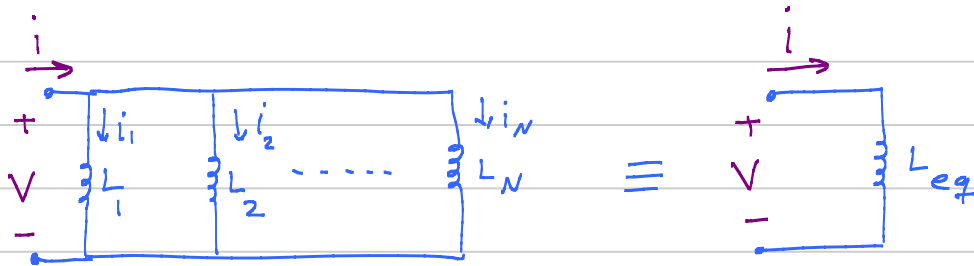
series combination of inductors



$$\begin{aligned} V &= V_1 + V_2 + \dots + V_N = L_1 \frac{di}{dt} + L_2 \frac{di}{dt} + \dots + L_N \frac{di}{dt} \\ &= \underbrace{(L_1 + L_2 + \dots + L_N)}_{L_{eq}} \frac{di}{dt} \end{aligned}$$

$$\Rightarrow \boxed{L_{eq} = L_1 + L_2 + \dots + L_N} \quad \text{series inductors}$$

parallel combination of inductors



$$i = i_1 + i_2 + \dots + i_N = \frac{1}{L_1} \int v dt + i_{o1} + \frac{1}{L_2} \int v dt + i_{o2} + \dots + \frac{1}{L_N} \int v dt + i_{oN}$$

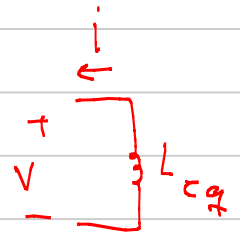
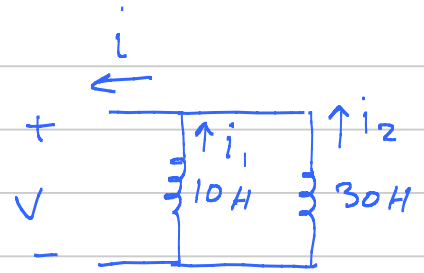
$$= \left(\frac{1}{L_1} + \frac{1}{L_2} + \dots + \frac{1}{L_N} \right) \int v dt + (i_{o1} + i_{o2} + \dots + i_{oN})$$

$$\Rightarrow \boxed{\frac{1}{L_{eq}} = \frac{1}{L_1} + \frac{1}{L_2} + \dots + \frac{1}{L_N} \quad , \quad i_{o,eq} = i_{o1} + i_{o2} + \dots + i_{oN}} \quad \text{Parallel inductors}$$

For two parallel inductors: $L_{eq} = \frac{L_1 L_2}{L_1 + L_2}$

Ex. If $v(t) = -1800 e^{-20t}$, $i_1(0) = 4A$ and $i_2(0) = -16A$,

- Find $i(t)$
- Find $i_1(t)$
- Find $i_2(t)$
- Find the initial energy stored in the parallel inductors.
- Find energy delivered to the source within $0 < t < \infty$.
- How much energy is trapped in the inductors?



$$a) L_{eq} = \frac{10 \times 30}{10 + 30} = 7.5 H, \quad i_{0eq} = 4 - 16 = -12 A$$

$$i = \frac{-1}{L_{eq}} \int_0^t v dt + i_{0eq} = \frac{-1}{7.5} \int_0^t -1800 e^{-20t} dt - 12$$

$$= \frac{1800}{7.5} \times \frac{-1}{20} (e^{-20t} - 1) - 12 = -12 (e^{-20t} - 1) - 12 = \underline{-12 e^{-20t} A}$$

$$b) i_1 = -\frac{1}{L_1} \int_0^t v dt + i_{01} = -\frac{1}{10} \times -1800 \times \frac{-1}{20} (e^{-20t} - 1) + 4$$

$$\underline{i_1 = -9 e^{-20t} + 13}$$

$$c) i_2 = i - i_1 = \underline{-3 e^{-20t} - 13}$$

$$d) W_0 = W_{01} + W_{02} = \frac{1}{2} L_1 i_{01}^2 + \frac{1}{2} L_2 i_{02}^2 = \frac{1}{2} (10) (4)^2 + \frac{1}{2} (30) (-16)^2 = \underline{3920 J}$$

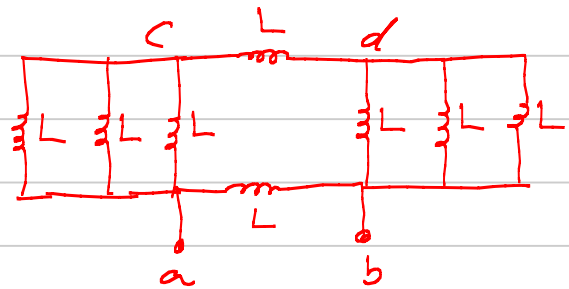
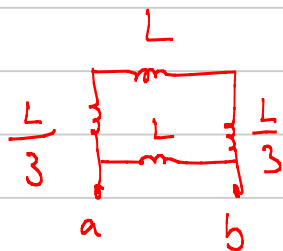
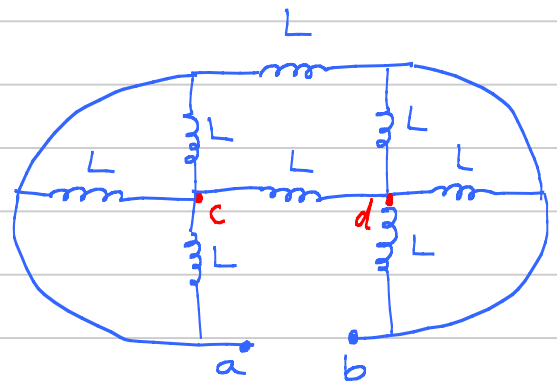
$$\begin{aligned}
 e) \quad W_{del.} &= W_s(\infty) - W_s(0) = \frac{1}{2} L_{eq} (i_{\infty})^2 - \frac{1}{2} L_{eq} (i_0)^2 \\
 &= \frac{1}{2} (7.5) (0)^2 - \frac{1}{2} (7.5) (-12)^2 = \underline{-540 \text{ J}}
 \end{aligned}$$

$$f) \quad W_{trapped} = 3920 - 540 = 3380 \text{ J}$$

$$\text{check: } \frac{1}{2} L_1 (i_{1\infty})^2 + \frac{1}{2} L_2 (i_{2\infty})^2 = 3380 \text{ J} \checkmark$$

Assuming ideal inductors (have zero resistance), there is no dissipative element in the circuit. Part of the initial energy stored in the inductors are delivered to the source and the remaining can be stored in the inductor forever if the inductors are ideal.

Ex. Find the equivalent inductance.



$$L_{eq} = \left(\frac{L}{3} + \frac{L}{3} + L \right) \parallel L = \underline{\underline{\frac{5}{8} L}}$$